

Alizée Amsler Moulanier, PhD

Post-doc in planetary science

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🌐 Alizée Amsler Moulanier 🆔 0009-0000-7492-1476 📄 Alizée Amsler Moulanier

PROFILE

I obtained my PhD in planetary science from the *Laboratoire d'Astrophysique de Marseille*. My work was funded by the *Institut Origines* of Aix-Marseille University. I focus on the study of the evolution of the icy moons' hydrospheres, by modelling the physical and chemical processes affecting the volatiles distribution. My work will contribute to the interpretation and exploitation of *JUICE* and *Europa Clipper* future data. I am affiliated to Europa Clipper-MASPEX instrument.

EDUCATION

10/2022 – 12/2025 Marseille, France	Ph.D in Astrophysics , Aix-Marseille Université, Laboratoire d'Astrophysique de Marseille 🔗 "Physico-chemical evolution of the icy moon's hydrospheres" Supervised by Pr. Olivier Mousis and Dr. Alexis Bouquet
09/2022 Milano, Italy	MSc Space Engineering , Politecnico Di Milano 🔗 Double-degree exchange. Focus on space mission design.
08/2020 Nantes, France	Diplôme d'Ingénieur généraliste , Ecole Centrale de Nantes 🔗 General engineering studies. Focus on bio-informatics
09/2016 – 07/2018 Lyon, France	Classe préparatoire aux grandes écoles , Lycée Aux Lazaristes Intense preparatory course of 2 years. Speciality in Physics and Chemistry

PROFESSIONAL EXPERIENCE

03/2026 – Present	University of Bern, Physics Institute Postdoctoral researcher part of Prof. Audrey Vorburger's group. Funded by the SNSF.
10/2022 – 12/2025 Marseille, France	Institut Origines, Aix-Marseille Université PhD student
03/2022 – 10/2022 Milano, Italy	Politecnico Di Milano Master thesis supervised by Pr. Mauro Massari : "Feasibility analysis of an astrochemistry payload for cubesat"
04/2020 – 08/2020 Evry, France	Genoscope - CEA 🔗 Research internship in a environmental genomics laboratory. Supervised by Dr. Eric Pelletier

TEACHING

03/2024 – 09/2024 Marseille, France	6 months Internship mentoring Mentoring of a Bachelor student on the chemical evolution of the icy moons.
01/2024 – 04/2024 Marseille, France	Science Watch mentoring Co-supervision of a master student for a <i>Science-Watch</i> project, focused on the <i>Delivery of organic matter to the icy moons' primordial hydrosphere</i> .
2023 – 2024 OHP, France	SIMO internship mentoring 40h teaching of masters' students on the use of T80 telescope and astronomical image processing.
Marseille, France	Aix-Marseille University's Master of Fundamental physics Short hour-long interventions in the planetary science course, for the 2nd year students of the Fundamental Physics Master.

PUBLIC OUTREACH AND ACADEMIC RESPONSABILITY

2024 Marseille, France	GSP coffee Hosting the weekly team journal club, discussing scientific news and papers.
09/2024 La Ciotat, France	Festival Lumexplore  Outreach presentation to a public audience : <i>Habitability of the Oceans Worlds</i>
01/2024 – 06/2024 Marseille, France	Apprentis Chercheurs  Mentoring of two 14 and 16 years old students in a short research project.
10/2023 Marseille, France	Fête de la Science Participation to the public event organized by the city of Marseille, to promote science to the public and share the lab's activities.
04/2023 Martigues, France	Pint of Science festival  Outreach presentation to a public audience : <i>Life in the Solar System</i>
03/2023 Marseille, France	Classroom' scientific partner Four interventions in classroom for 7-8 years old children.
2023 – 2025 Marseille, France	Astrogirls  Participations to the <i>Astrogirls</i> days to promote women in astrophysics.

COURSES

06/2024 OHP, France	Sample Returns Summer School  Participant
06/2023 OHP, France	Moons of the Solar System Summer School  Participant and LOC member
02/2023	GWB Workshop 3 days introduction workshop to the <i>Geochemist Workbench</i> .

SKILLS

- Python	- Matlab	- Geochemist Workbench
- Inkscape	- WORM	- PHREEQ-C

LANGUAGES

French Mother tongue	English Fluent	German Father tongue	Italian Beginner
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PUBLICATIONS

06/2025	Noble gas depletion on Titan: clathrate sequestration during the open ocean phase , Astronomy & Astrophysics Alizée Amsler Moulanier , Olivier Mousis, Alexis Bouquet and Ngan H.D. Trinh https://doi.org/10.1051/0004-6361/202554398 
01/2025	The role of ammonia in the distribution of volatiles in the primordial hydrosphere of Europa  , Planetary Science Journal Alizée Amsler Moulanier , Olivier Mousis, Alexis Bouquet and Christopher R. Glein https://doi.org/10.3847/PSJ/ad9925 
11/2025	Delivery of organic matter to the galilean moons , MNRAS Tom Benest Couzinou, Alizée Amsler Moulanier and Olivier Mousis https://doi.org/10.1093/mnras/staf2074 
2026	On the chemical evolution of Europa's hydrosphere during the accretion phase , In prep Alizée Amsler Moulanier , Olivier Mousis, Yasuhito Sekine, Yannis Bennacer, Shintaro Kadoya

CONFERENCE AND WORKSHOP ATTENDANCE

Helsinki, Finland	EPSC 2025 , Poster presentations
Vienna, Austria	EGU 2025 , Oral presentation

Berlin, Germany	EPSC 2024 ↗ , Oral and Poster presentations
Marseille, France	SF2A 2024 ↗ , Oral presentation
Vienna, Austria	EGU 2024 ↗ , Oral presentation
Göttigen, Germany	JUICE-SWI 2024 April science meeting , Oral presentation
Toulouse, France	ANR FACOM 2023 December science meeting , Two oral presentations
Göttigen, Germany	JUICE-SWI 2023 November science meeting , Oral presentation
Marseille, France	Workshop on the Origins and Habitability of the Galilean Moons ↗ , Poster presentation, LOC member
Marseille, France	International Planetary Probe Workshop 2023 ↗ , Poster presentation, LOC member
La Palma, Spain	BEACON 2023 ↗ , Poster presentation
Marseille, France	ANR FACOM January 2023 science meeting , Oral presentation